

Unit 21

ORGANIC SYNTHESIS

10 items · 39 marks

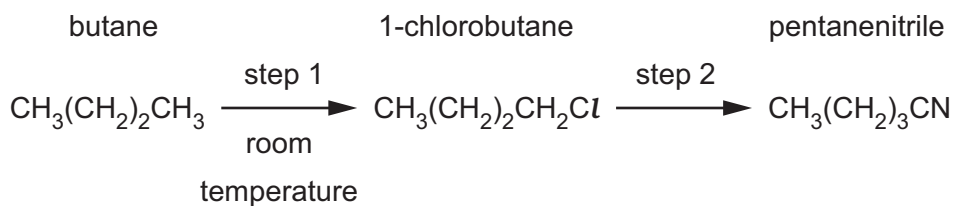
Syllabus sections in this unit

21.1 Organic synthesis

Unit 21: Organic synthesis**1. M/J 26/P24/Q4(d)(i)**

Context: part (d) (introduction)

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

**Figure 4.3**

(i) Suggest the reagent and conditions for each step in Figure 4.3.

step 1

step 2

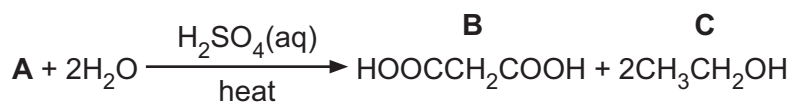
[2]

2. M/J 25/P21/Q5(d)(i)

Context

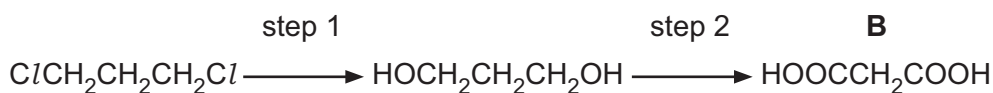
5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (d) (introduction)

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

**Fig. 5.2**

(i) Identify the reagents and conditions required in steps 1 and 2.

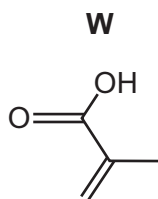
step 1

step 2

[2]

3. M/J 25/P23/Q5(a)(ii)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

(ii) Two different reagents are each added to separate samples of **W** as shown in Table 5.1.

Complete Table 5.1.

Table 5.1

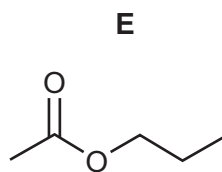
reagent and conditions	observation when reagent is added to W	structural formula of organic product when reagent is added to W
solid Na		
Br ₂ in absence of ultraviolet light		

[4]

4. M/J 25/P24/Q4(d)(iii)

Context

4 The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

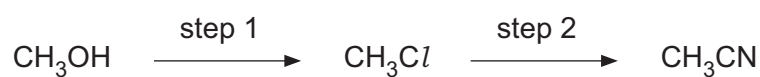
(d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

(iii) CH_3OH is used to make CH_3CN in a two-step process, as shown in Fig. 4.2.

**Fig. 4.2**

State the reagents and conditions required for steps 1 and 2.

step 1

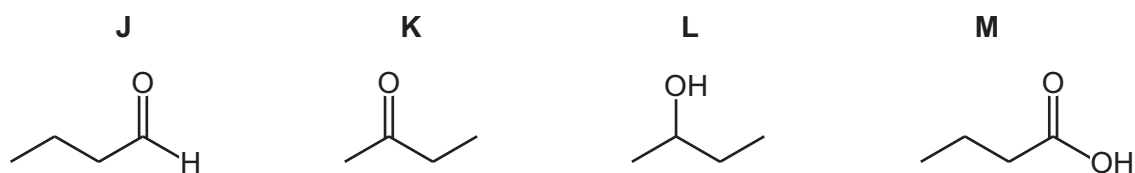
step 2

[3]

5. MAR 25/P22/Q4(a)

Context

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

(a) Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$		J and L
alkaline $\text{I}_2(\text{aq})$	yellow precipitate	
	orange precipitate	J and K
$\text{Na}(\text{s})$		

[5]

6. M/J 24/P22/Q4(c)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(c) Separate samples of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$, $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ and $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$ are tested with different reagents.

Complete Table 4.1. If no reaction occurs, write $\times$ in the relevant box.

Table 4.1

reagent added	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
$\text{Br}_2(\text{l})$ in the dark			
$\text{PCl}_5(\text{s})$			
$\text{AgNO}_3(\text{aq})$			

[3]



7. O/N 23/P22/Q4(b)(iii)

Context

- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

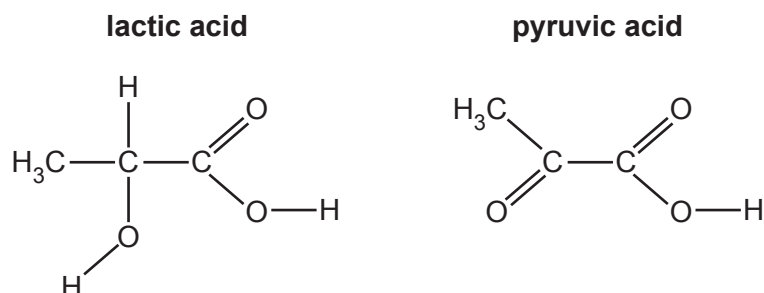


Fig. 4.1

Context: part (b) (introduction)

- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

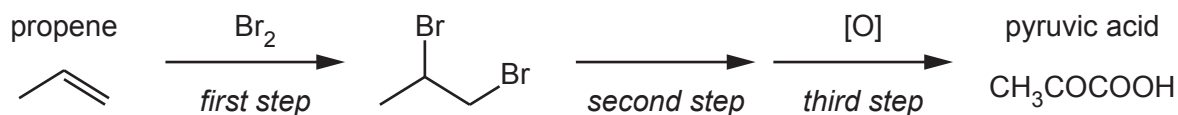


Fig. 4.3

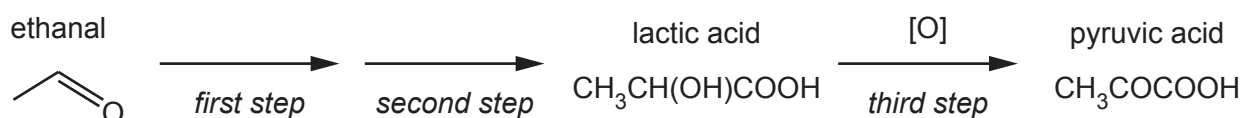


Fig. 4.4

- (iii) Complete Table 4.1 to give details of the reagents and conditions used in each of the two syntheses shown in Fig. 4.3 and Fig. 4.4.

Table 4.1

		synthesis from propene (shown in Fig. 4.3)	synthesis from ethanal (shown in Fig. 4.4)
reagents and conditions used	first step	Br_2	
	second step		
	third step		

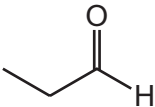
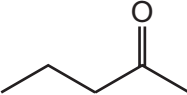
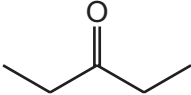
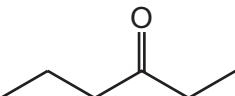
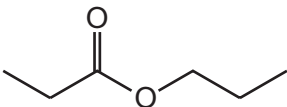
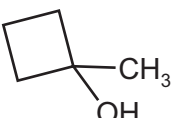
8. O/N 22/P22/Q3(a)(i)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

Context: part (a) (introduction)

(a) Table 3.1 shows four pairs of organic compounds.**Table 3.1**

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(i) Complete Table 3.1 to:

- identify a reagent which can distinguish between the compounds in each pair
- give the **positive** result of the chemical test **and** identify which compound shows this result.

Use a different reagent for each test.

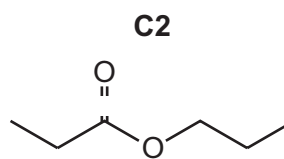
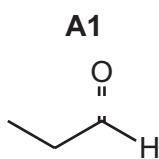
[8]

9. O/N 22/P22/Q3(d)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(d) **C2** can be synthesised using **A1** as a single organic reactant.



Devise a multi-step synthetic route to form **C2** from **A1**.
Identify relevant reagents and conditions, and state the organic products of each step.

[3]

10. MAR 22/P22/Q4(b)(iv)

Context

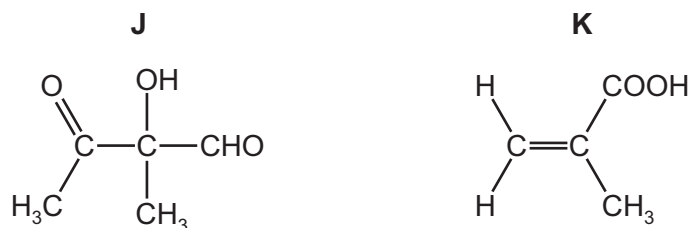
4 Compounds **J** and **K** are found in plant oils.

Fig. 4.1

Context: part (b) (introduction)

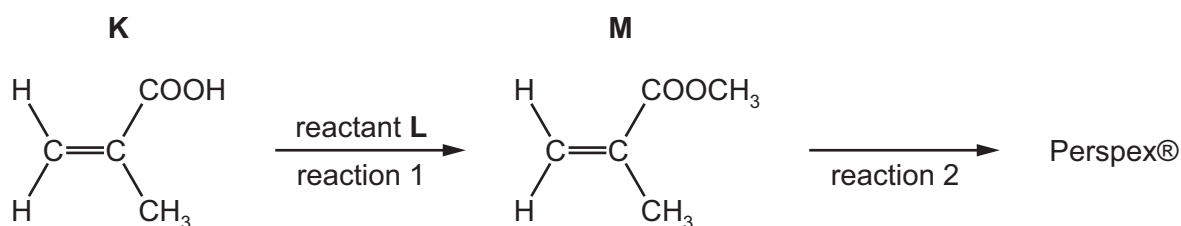
(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

Fig. 4.2

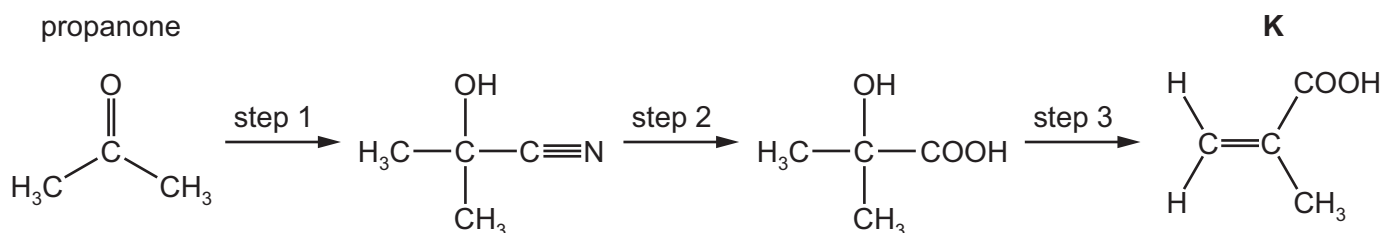
(iv) **K** can be made from propanone in the three-step synthesis shown in Fig. 4.3.

Fig. 4.3

Complete Table 4.3 to identify the reagent(s) used and the type of reaction in each step.

Table 4.3

step	reagent(s)	type of reaction
1		
2		
3	Al_2O_3	

Answers Section

1. M/J 26/P24/Q4(d)(i)

4(d)(i)	M1 (step 1) Cl_2 AND ultraviolet (light) / UV (light) M2 (step 2) KCN in ethanol AND heat (under reflux)	2	M2 A ethanolic KCN AND heat alcohol = (alc) A NaCN in ethanol AND heat R HCN R KCN in (aq) ethanol R ethanoic I warm high temp = heat = reflux = $> 60^\circ\text{C}$
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2. M/J 25/P21/Q5(d)(i)

5(d)(i)	M1 NaOH(aq) AND heat M2 acidified AND potassium dichromate(VI) AND heat under reflux	2
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3. M/J 25/P23/Q5(a)(ii)

5(a)(ii)	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%;">reagent</th><th style="width: 35%;">observation when reagent is added to W</th><th style="width: 50%;">structural formula of organic product when reagent is added to W</th></tr> </thead> <tbody> <tr> <td>Na</td><td>fizz / bubbles etc / OR solid / Na disappears / dissolves</td><td>$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$</td></tr> <tr> <td>$\text{Br}_2$ in the dark</td><td>brown to colourless / decolourise</td><td>$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$</td></tr> </tbody> </table>	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$	Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$	4
reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W									
Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$									
Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$									

4. M/J 25/P24/Q4(d)(iii)

4(d)(iii)	M1 step 1 PCl_3 and heat OR $\text{HCl}(\text{g})$ OR PCl_5 OR SOCl_2 OR NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 M2 step 2 reagent KCN M3 conditions for step 2 dissolved in ethanol AND heat	3
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5. MAR 25/P22/Q4(a)

4(a)	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 30%;">reagent</th><th style="width: 30%;">observation</th><th style="width: 40%;">conclusion</th></tr> </thead> <tbody> <tr> <td>acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$</td><td>(solution) turns (from orange to) green</td><td>J and L</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>yellow precipitate forms</td><td>K and L</td></tr> <tr> <td>2,4-dinitrophenylhydrazine / 2,4-DNPH</td><td>orange precipitate forms</td><td>J and K</td></tr> <tr> <td>Na(s)</td><td>effervescence</td><td>L and M</td></tr> </tbody> </table>	reagent	observation	conclusion	acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L	alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L	2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K	Na(s)	effervescence	L and M	5
reagent	observation	conclusion															
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L															
alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L															
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K															
Na(s)	effervescence	L and M															

6. M/J 24/P22/Q4(c)

4(c)

Chemical test	observation with 2-bromooctane	observation with octan-2-ol	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
Br_2 in the dark	✗		orange / brown to colourless
PCl_5		steamy fumes	✗
$\text{AgNO}_3(\text{aq})$	cream precipitate OR off white precipitate	✗	

6 correct answers = 3 marks
4 or 5 correct answers = 2 marks
2 or 3 correct answers = 1 mark
0 or 1 correct answer = 0 marks

3

7. O/N 23/P22/Q4(b)(iii)

4(b)(iii)	$\text{HCN} \& \text{KCN}$ OR NaCN with H_2SO_4	1
	$\text{NaOH}(\text{aq})$ (1) dilute / aq(ueous) HCl / H_2SO_4 (1)	2
	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	1

8. O/N 22/P22/Q3(a)(i)

3(a)(i)	M1 Tollens' OR Fehling's OR $\text{K}_2\text{Cr}_2\text{O}_7$ OR KMnO_4 OR alkaline $\text{I}_2(\text{aq})$ with	M2 A1 gives silver mirror OR brick-red ppt OR (orange solution) turns green OR (purple solution) turns colourless A2 gives yellow ppt	2
	M3 alkaline $\text{I}_2(\text{aq})$	M4 B1 gives yellow ppt	2
	M5 2,4-DNPH	M6 C1 gives red/orange/yellow ppt	2
	M7 $\text{Br}_2(\text{aq})$	M8 D2 turns it (orange to) colourless	2

9. O/N 22/P22/Q3(d)

3(d)	M1 one portion of A1 : NaBH_4 AND $\rightarrow$ propan-1-ol / $\text{CH}_3(\text{CH}_2)_2\text{OH}$	1
	M2 second portion of A1 : acidified $\text{K}_2\text{Cr}_2\text{O}_7$ AND $\rightarrow$ propanoic acid / $\text{C}_2\text{H}_5\text{CO}_2\text{H}$	1
	M3 two products: <u>conc</u> (entrated mineral) acid / H^+ / H_2SO_4 (catalyst)	1

10. MAR 22/P22/Q4(b)(iv)

4(b)(iv)	step 1 KCN / HCN OR $\text{NaCN} / \text{H}_2\text{SO}_4$ [1] step 2 $\text{H}^+ / \text{H}_2\text{SO}_4(\text{aq})$ [1] step 3	addition [1] hydrolysis / substitution [1] elimination / dehydration [1]	5
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